

Global Radial Monotonicity for Harmonic Polylogarithms

A uniform outer-order theorem, quantitative
inner-order stability, and positive binomial identities

Research continuation prepared for
Vladimir Reshetnikov's ProveIt project

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Abstract. For $F_{a,b}(z) = \sum_{n>m\geq 1} z^n / (n^a m^b)$, let $\theta_{a,b}(\rho)$ be the unique upper-semicircle zero of its imaginary part and put $\eta_{a,b}(\rho) = \cos \theta_{a,b}(\rho) / \rho$. We prove strict decrease of η on the entire interval $0 < \rho \leq 1$ for every real $a \geq 8$ and every $b > 0$, with the explicit differential bound $\eta'(\rho) < -\rho H_4^{(b)}(2/5)^a / 120$. The argument combines a uniform localization of the zero, a low-mode inequality certified by ten positive rational Bernstein coefficients, and analytic bounds for the full infinite tail. A separate stability theorem proves the same global direction for every $a > 0$ whenever $2^{-b} < [(2/5)^a - (3/8)^a] / 256$. Thus the integer outer-order conjecture is reduced to six bounded inner-order strips, rather than merely verified near the origin. We derive a weighted cubic-moment criterion for radial motion, a positive squared-resolvent representation, and convergent binomial identities with exact remainders, including a specialization for $\pi \log 2$. Standard-library rational verifiers certify the uniform bounds, eight angular-root brackets, and nine Gaussian values. The full normalized-radius conjecture and the arithmetic S_6, S_8 candidates are not claimed resolved.

Ordinary analytic proofs with finite exact certificates; not a proof-assistant formalization or an externally refereed paper. No literature-wide priority claim is made.

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1 Research target, provenance, and principal results

The current manuscript *Polylogarithms and their Arithmetic Bridges* already proves that the first nonconstant radial coefficient has the conjectured sign for all real outer orders $a \geq 2$ and all positive inner orders b [1]. That is a local theorem: it does not exclude a derivative changing sign farther along the zero curve. The manuscript also proves global decrease in the exact limit $b = \infty$, while explicitly declining to infer the same assertion for finite b [2]. These are the two gaps addressed here.

Our repository reference is the retrieved revision

3338e535ef54ad8f6be56bdce8b3d084a2924465.

The relevant chapter paths and blob identifiers are recorded in `provenance/SOURCES.md`. The incoming directory was inventoried; this is not a claim that every incoming archive was fully audited. The accessible incoming quartic-bifurcation article treats a distinct, local fractional-order transition and two turning points [5]. The incoming all-depth and sharp-Euler articles likewise address different questions [6, 7].

1.1 Conventions and inherited facts

For $a, b > 0$, write

$$F_{a,b}(z) = \sum_{n \geq 2} c_n z^n, \quad c_n = \frac{H_{n-1}^{(b)}}{n^a}, \quad H_k^{(b)} = \sum_{j=1}^k j^{-b}. \quad (1.1)$$

The sum is strict, not a multiple-star sum. We use the principal continuation to $\mathbb{C} \setminus [1, \infty)$, consistent with the depth-one conventions in DLMF §25.12 [8]. Every boundary value below lies away from the cut. In both parameter regimes of our global theorems, $a + b > 1$, so the boundary series is also ordinarily convergent.

The manuscript's finite signed-measure theorem supplies the following fact on the domain $a, b > 0, a + b > 1$: on every open upper semicircle $z = \rho e^{i\theta}$, $0 < \rho \leq 1$, the imaginary part has exactly one simple zero, denoted $\theta_{a,b}(\rho)$, with

$$\arccos \rho < \theta_{a,b}(\rho) < \frac{\pi}{2}, \quad \partial_\theta \operatorname{Im} F_{a,b}(\rho e^{i\theta}) < 0 \quad \text{at the zero.} \quad (1.2)$$

This statement and its proof are inherited from [4], not claimed as a new result. For $a \geq 8$ we also give an independent uniqueness argument using the same estimates as the radial proof. Define

$$s = \rho^2, \quad m(s) = \frac{2 \cos \theta_{a,b}(\rho)}{\rho}, \quad \eta_{a,b}(\rho) = \frac{m(\rho^2)}{2}. \quad (1.3)$$

In particular, $\eta'(\rho) = \rho m'(s)$; this factor is essential.

1.2 Statements of the two global results

Theorem 1.1 (Uniform outer-order theorem). *For every real $a \geq 8$, every $b > 0$, and $0 < \rho \leq 1$,*

$$\eta'_{a,b}(\rho) < -\frac{\rho}{120} H_4^{(b)} \left(\frac{2}{5} \right)^a < 0. \quad (1.4)$$

The function extends continuously to zero with

$$\eta_{a,b}(0) = \frac{1 + 2^{-b}}{2} \left(\frac{2}{3}\right)^a, \quad (1.5)$$

and consequently

$$\eta_{a,b}(0) - \eta_{a,b}(\rho) > \frac{\rho^2}{240} H_4^{(b)} \left(\frac{2}{5}\right)^a. \quad (1.6)$$

At $\rho = 1$, the derivative is the derivative of the local analytic continuation of the zero curve, equivalently its left derivative.

Theorem 1.2 (Quantitative finite-inner-order transfer). For $a > 0$, put

$$\Delta(a) = \left(\frac{2}{5}\right)^a - \left(\frac{3}{8}\right)^a > 0. \quad (1.7)$$

If $b > 0$ satisfies

$$2^{-b} < \frac{\Delta(a)}{256}, \quad (1.8)$$

then $\eta'_{a,b}(\rho) < 0$ for every $0 < \rho \leq 1$. Under the stronger condition $2^{-b} \leq \Delta(a)/512$,

$$\eta'_{a,b}(\rho) \leq -\frac{\rho \Delta(a)}{16}. \quad (1.9)$$

For integer a the first sufficient integer thresholds in (1.8) are as follows; these are not optimal thresholds for the actual phenomenon.

Outer order a	1	2	3	4	5	6	7	8
Sufficient $b \geq$	14	14	15	16	17	18	19	20

For $a = 8$ the entry is redundant by [Theorem 1.1](#). Thus the original integer conjecture for $a \geq 2$ is reduced to the six strips $2 \leq a \leq 7$, with the integer a fixed and $0 < b < B_a$ bounded by this table. No proof is claimed for every point of those remaining strips.

1.3 Scope and originality discipline

The additions relative to the inspected sources are the two explicit all-radius regimes, their quantitative gaps, the cubic-moment velocity formula, and the stated binomial identities and certificates. Positive integral manipulations, Chebyshev representations, Bernstein positivity, and elementary geometric acceleration are background methods. An identity derived here need not be absent from the wider literature. In particular, the new positive series is not asserted to improve the sharp universal Euler evaluator in the incoming work.

2 Positive kernels, moments, and a real zero equation

2.1 The inherited double kernel and its probability form

The manuscript proves [\[3\]](#)

$$F_{a,b}(z) = z^2 \int_{0 < y < x < 1} \frac{W_{a,b}(x, y)}{(1 - xz)(1 - yz)} dy dx, \quad (2.1)$$

where

$$W_{a,b}(x, y) = \frac{(-\log x)^{a-1}(\log(x/y))^{b-1}}{\Gamma(a)\Gamma(b)}, \quad \int W_{a,b} = 2^{-a} = c_2. \quad (2.2)$$

For completeness, the substitution $y = xv$ gives

$$\int W_{a,b}(x, y)x^j y^k dy dx = \frac{1}{(j+k+2)^a(k+1)^b}.$$

Geometric expansion and summation over $j+k = n-2$ recover c_n . Uniform domination on compact subsets of $\mathbb{C} \setminus [1, \infty)$ proves continuation. The positivity of this kernel will be used, but does not by itself settle the sign of a normalized radial derivative.

Let Y, Z be independent gamma variables with shapes a, b and rates 2, 1, respectively. Set $X = e^{-Y}$ and $V = e^{-Z}$. The same substitution rewrites (2.1) as

$$F_{a,b}(z) = c_2 z^2 \mathbb{E} \frac{1}{(1-Xz)(1-XVz)}, \quad \mathbb{E} X^k = \left(\frac{2}{k+2} \right)^a, \quad \mathbb{E} V^k = (k+1)^{-b}. \quad (2.3)$$

All variables lie strictly between zero and one.

2.2 A positive squared-resolvent measure

Proposition 2.1. *There is a positive finite measure $\mu = \mu_{a,b}$, with full support $[0, 1]$ and mass c_2 , such that*

$$F_{a,b}(z) = z^2 \int_0^1 \frac{d\mu(u)}{(1-uz)^2}, \quad \int_0^1 u^k d\mu(u) = \frac{c_{k+2}}{k+1} \quad (k \geq 0). \quad (2.4)$$

In particular, for $\ell, k \geq 0$,

$$\sum_{j=0}^k (-1)^j \binom{k}{j} \frac{H_{\ell+j+1}^{(b)}}{(\ell+j+1)(\ell+j+2)^a} > 0. \quad (2.5)$$

Every finite Hankel moment matrix of μ is positive definite.

Proof. Use the elementary identity

$$\frac{1}{(1-xz)(1-yz)} = \int_0^1 \frac{dt}{[1-(tx+(1-t)y)z]^2}$$

in (2.1). The image of the positive product measure under $u = tx + (1-t)y$ defines μ . Every nonempty open subinterval of $(0, 1)$ has positive mass, so the support is full. Comparing coefficients in $(1-uz)^{-2} = \sum_{k \geq 0} (k+1)u^k z^k$ proves the moments. The left side of (2.5) is $\int u^\ell (1-u)^k d\mu > 0$. For a nonzero polynomial p , $\int p(u)^2 d\mu(u) > 0$, proving the Hankel assertion. $\square$

This is a positive representation with a *squared* resolvent. It does not contradict the failure of a finite signed *first*-resolvent representation in part of the low-order domain.

2.3 Two forms of the zero equation

For $s = \rho^2$ and $m = 2 \cos \theta / \rho$ let

$$D_u(m, s) = 1 + s(u^2 - mu). \quad (2.6)$$

Rationalizing (2.4) yields

$$\operatorname{Im} F_{a,b}(\rho e^{i\theta}) = \rho^3 \sin \theta P(m, s), \quad P(m, s) = \int \frac{m - 2u}{D_u(m, s)^2} d\mu(u). \quad (2.7)$$

Similarly (2.3) yields

$$P(m, s) = c_2 G_b(m, s), \quad G_b(m, s) = \mathbb{E} \frac{m - X - XV}{D_X(m, s) D_{XV}(m, s)}. \quad (2.8)$$

In particular $P(0, s) < 0$ for all $0 \leq s \leq 1$, and the imaginary part is negative at every upper-half-disk point with $\cos \theta \leq 0$.

For the series version, let U_n denote the Chebyshev polynomial of the second kind. Its standard trigonometric representation is $U_{n-1}(\cos \theta) = \sin(n\theta)/\sin \theta$ [9]. Define

$$T_n(m, s) = \rho^{n-3} U_{n-1}(\rho m/2). \quad (2.9)$$

Parity shows that this is a polynomial in m, s for $n \geq 2$, and

$$T_2 = m, \quad T_3 = sm^2 - 1, \quad T_{n+1} = s(mT_n - T_{n-1}), \quad P(m, s) = \sum_{n \geq 2} c_n T_n(m, s). \quad (2.10)$$

The next four polynomials are

$$\begin{aligned} T_4 &= s^2 m^3 - 2sm, & T_5 &= s^3 m^4 - 3s^2 m^2 + s, \\ T_6 &= s^4 m^5 - 4s^3 m^3 + 3s^2 m, & T_7 &= s^5 m^6 - 5s^4 m^4 + 6s^3 m^2 - s^2. \end{aligned} \quad (2.11)$$

At $s = 0$ the equation is $c_2 m - c_3 = 0$, giving $m(0) = c_3/c_2$ and (1.5).

3 Uniform localization and control of the angular derivative

From now until the end of section 4, assume $a \geq 8, b > 0$. Write

$$x = 2^{-b}, \quad y = 3^{-b}, \quad U = 1 + x, \quad V_0 = 1 + x + y, \quad W = 1 + x + y + x^2. \quad (3.1)$$

We distinguish V_0 from the random variable V in (2.3). Then $c_3 = U3^{-a}, c_4 = V_0 4^{-a}, c_5 = W5^{-a}$, and

$$0 < x < 1, \quad x^2 \leq y \leq x, \quad \frac{U^2}{V_0} \leq \frac{4}{3}. \quad (3.2)$$

The last inequality follows from $(1+x)^2/(1+x+x^2) \leq 4/3$.

Lemma 3.1 (Uniform coefficient comparisons). *For the indicated ranges of n ,*

$$\begin{aligned} \frac{c_n}{c_2} &\leq (n-1) \left(\frac{2}{n}\right)^a & (n \geq 2), \\ \frac{c_n}{c_3} &\leq \frac{n-1}{2} \left(\frac{3}{n}\right)^a & (n \geq 3), \\ \frac{c_n}{c_5} &\leq \frac{n-1}{4} \left(\frac{5}{n}\right)^a & (n \geq 5). \end{aligned} \quad (3.3)$$

Proof. The averages $H_k^{(b)}/k$ decrease because the summands j^{-b} decrease. Compare these averages at $k = n-1$ and at $k = 1, 2, 4$. $\square$

Lemma 3.2 (Elementary Chebyshev bounds). *For $n \geq 3$ and $0 \leq c \leq 1/6$,*

$$|U'_{n-1}(c)| \leq \frac{11}{10}n. \quad (3.4)$$

For $0 \leq c \leq 1/20$,

$$|U_{n-1}(c)| \leq \begin{cases} 101/100, & n \text{ odd}, \\ (101/100)nc, & n \text{ even}. \end{cases} \quad (3.5)$$

For all $-1 \leq c \leq 1$, $|U_{n-1}(c)| \leq n$.

Proof. Differentiating the trigonometric representation gives

$$|U'_{n-1}(c)| \leq \frac{n}{1-c^2} + \frac{c}{(1-c^2)^{3/2}} \leq \frac{36n}{35} + \frac{1}{5} < \frac{11n}{10}.$$

Here $n \geq 3$ is used in the final comparison. Write $\theta = \pi/2 - \delta$, so $\sin \delta = c$. For odd n , use $|\sin(n\theta)| \leq 1$; for even n , use $|\sin(n\delta)| \leq n \sin \delta$. Division by $\sin \theta = \sqrt{1-c^2} > 100/101$ gives (3.5). Finally $|\sin(n\theta)| \leq n \sin \theta$ proves the global bound, including endpoints by continuity. $\square$

Set

$$C = \frac{11}{10}, \quad q = \frac{c_3}{c_2}, \quad M = Cq, \quad M_8 = \frac{11}{5} \left(\frac{2}{3}\right)^8 = \frac{2816}{32805} < \frac{1}{10}. \quad (3.6)$$

Three rationally certified bounds used below are

$$\begin{aligned} B_8 &:= \sum_{n \geq 3} n(n-1)(2/n)^8 < \frac{1}{3}, \\ C \left(1 - \frac{101}{200}E_8\right) - \frac{101}{100}O_8 &> \frac{1}{50}, \\ O_8 &:= \sum_{\substack{n \geq 3 \\ n \text{ odd}}} \frac{n-1}{2}(3/n)^8, \quad E_8 := \sum_{\substack{n \geq 4 \\ n \text{ even}}} n(n-1)(2/n)^8. \end{aligned} \quad (3.7)$$

The exact finite-sum and integral-tail verification is given in [section A](#); no decimal approximation is an acceptance test.

Proposition 3.3 (Root strip and derivative bounds). *Every upper-semicircle zero lies in*

$$0 < m(s) < C \frac{c_3}{c_2} \leq M_8 < \frac{1}{10}. \quad (3.8)$$

It is the only such zero, is simple, and on the larger strip $0 \leq m \leq 1/3, 0 \leq s \leq 1$,

$$\frac{4}{5}c_2 < P_m(m, s) < \frac{6}{5}c_2. \quad (3.9)$$

Proof. There is no zero with $\cos \theta \leq 0$ by (2.8). For any zero with $m > 0$, the global Chebyshev bound and (3.3) give

$$c_2 m \leq \sum_{n \geq 3} n c_n \rho^{n-3} \leq c_2 B_8,$$

so all possible zeros lie in $m < 1/3$. On this strip $c = \rho m/2 \leq 1/6$ and $\partial_m T_n = \rho^{n-2} U'_{n-1}(c)/2$. Therefore

$$|P_m - c_2| \leq \frac{11}{20} \sum_{n \geq 3} n c_n \leq \frac{11}{20} B_8 c_2 < \frac{11}{60} c_2,$$

which implies (3.9).

We have $P(0, s) < 0$. At $m = M$, the parity bounds apply because $M \leq M_8 < 1/10$. Bound all odd-indexed terms $n \geq 3$ and all even-indexed terms $n \geq 4$ in absolute value, while retaining $c_2 M$:

$$\frac{P(M, s)}{c_3} \geq C \left(1 - \frac{101}{200} E_8 \right) - \frac{101}{100} O_8 > \frac{1}{50}.$$

The coefficient comparisons and the inequalities $(r/n)^a \leq (r/n)^8$ for $n \geq r$ justify the uniform replacements. Existence follows by the intermediate value theorem, and strict increase of P on the containing strip proves uniqueness and simplicity. The estimates also justify termwise differentiation: the relevant majorants are bounded by constant multiples of $\sum n^{2-a}$. At zero the implicit-function theorem applies to $P(m, 0) = c_2 m - c_3$. Away from zero, analyticity of F off the cut and simplicity give the local analytic continuation, including through $s = 1$. $\square$

4 The uniform radial derivative inequality

We now prove the main sign. The distinction between localization and radial monotonicity matters: a small zero does not by itself fix the sign of its velocity.

4.1 A positive low-mode contribution

Lemma 4.1 (The first five relevant modes). *At every point with $0 \leq m \leq M, 0 \leq s \leq 1$,*

$$\sum_{n=2}^6 c_n \partial_s T_n \geq c_5 + c_3 m^2 - 2c_4 m - 6c_5 m^2 > \frac{2}{5} c_5. \quad (4.1)$$

Proof. From (2.11),

$$\partial_s T_3 = m^2, \quad \partial_s T_4 \geq -2m, \quad \partial_s T_5 \geq 1 - 6m^2,$$

and

$$\partial_s T_6 = sm(6 - 12sm^2 + 4s^2m^4) \geq 0$$

because $m < 1/10$. This proves the first inequality. Moreover

$$\frac{M}{c_4/c_3} = C \frac{U^2}{V_0} \left(\frac{8}{9}\right)^a \leq 2C \left(\frac{8}{9}\right)^8 < 1.$$

Thus $c_3m^2 - 2c_4m$ decreases on $[0, M]$. Substitution of $M = Cc_3/c_2$ and $m^2 \leq M^2$ gives a lower bound $c_5\mathcal{E}(a, b)$, where

$$\mathcal{E}(a, b) = 1 - 2C \frac{UV_0}{W} \left(\frac{5}{6}\right)^a + C^2 \frac{U^3}{W} \left(\frac{20}{27}\right)^a - 6C^2 U^2 \left(\frac{4}{9}\right)^a. \quad (4.2)$$

We prove that $\mathcal{E}(a, b) > 2/5$ uniformly.

First, $\mathcal{E}(a, b)$ increases with $a \geq 8$. For the two middle exponentials, the ratio of the magnitude of the negative derivative to the positive derivative is

$$\frac{CU^2}{2V_0} \left(\frac{8}{9}\right)^a \frac{\log(27/20)}{\log(6/5)} < \frac{22}{15} \left(\frac{8}{9}\right)^8 < 1.$$

We used $U^2/V_0 \leq 4/3$ and $27/20 < (6/5)^2$, which implies the logarithm ratio is less than two. The last term in (4.2) also has positive derivative in a .

It remains to verify $\mathcal{E}(8, b) > 2/5$. Multiply the difference by the positive W . The resulting polynomial is

$$p(x, y) = \frac{3}{5}W - 2C(5/6)^8 UV_0 + C^2(20/27)^8 U^3 - 6C^2(4/9)^8 U^2 W. \quad (4.3)$$

It is affine in y . On the entire larger region $0 \leq x \leq 1, x^2 \leq y \leq x$, positivity therefore follows from positivity at $y = x^2$ and $y = x$. Each endpoint polynomial has degree at most four. In the degree-four Bernstein basis $\binom{4}{j} x^j (1-x)^{4-j}$, all ten coefficients are strictly positive. Their exact numerators with a common denominator are in [table 1](#). This proves positivity on the full region, including the actual curve $y = 3^{-b}$, and finishes the proof. $\square$

Table 1: Exact Bernstein certificate for (4.3). Divide every displayed integer by 7230196133913600.

j	$p(x, x^2)$	$p(x, x)$
0	1351861016318884	1351861016318884
1	1121529834868342	1261248688948063
2	1434025919811540	977124901564326
3	1462870427049383	617821376075897
4	221755156299224	221755156299224

4.2 A bound for the entire remaining tail

Lemma 4.2 (Radial tail). *For $0 \leq m \leq M_8$ and $0 \leq s \leq 1$,*

$$\left| \sum_{n \geq 7} c_n \partial_s T_n(m, s) \right| < \frac{39}{100} c_5. \quad (4.4)$$

Proof. For $s > 0$, differentiation of (2.9) gives

$$\partial_s T_n = \frac{n-3}{2} \rho^{n-5} U_{n-1}(\rho m/2) + \frac{m}{4} \rho^{n-4} U'_{n-1}(\rho m/2).$$

Using Lemma 3.2 and $\rho \leq 1$ yields

$$|\partial_s T_n| \leq B_n := \begin{cases} \frac{101}{200}(n-3) + \frac{11}{40} M_8 n, & n \text{ odd}, \\ \frac{M_8 n(101n-193)}{400}, & n \text{ even}. \end{cases} \quad (4.5)$$

The even bound retains the extra factor $c = \rho m/2$ in (3.5); losing that factor would make the estimate too weak. By continuity the same bounds hold at $s = 0$. Now Lemma 3.1 bounds the tail divided by c_5 by

$$R_8 = \sum_{n \geq 7} \frac{n-1}{4} \left(\frac{5}{n} \right)^8 B_n. \quad (4.6)$$

Exact summation through $n = 100$ plus the integral bounds in section A gives

$$R_8 < 0.388094585988858 < \frac{39}{100}.$$

The displayed decimal is only a readable outward bound; the replay compares rational numbers exactly. The same majorant proves uniform convergence of the differentiated tail. $\square$

Proof of Theorem 1.1. At the zero, Lemmas 4.1 and 4.2 give

$$P_s(m(s), s) > \left(\frac{2}{5} - \frac{39}{100} \right) c_5 = \frac{c_5}{100} > 0. \quad (4.7)$$

Implicit differentiation and (3.9) imply

$$m'(s) = -\frac{P_s}{P_m} < -\frac{c_5}{120c_2}.$$

Using (1.3) proves (1.4). The equation at $s = 0$ gives (1.5); integration from zero to ρ proves (1.6). Every estimate is uniform in $b > 0$ and in the full interval $0 \leq s \leq 1$. $\square$

Remark 4.3 (Why this is not a local Taylor argument). Only finitely many modes are treated sharply, but the proof is not a truncation without an error estimate. The inequality $P_s > c_5/100$ incorporates *every* mode $n \geq 7$ and holds at *every* radius up to one. The Bernstein step retains the dependence between U, V_0, W ; replacing them by unrelated extreme values destroys the useful positive margin.

5 Quantitative transfer from the infinite-inner-order family

The limiting function is $F_{a,\infty}(z) = \text{Li}_a(z) - z$, whose strict global decrease is already proved in the manuscript [2]. Here we make that limit stable with an explicit finite- b criterion.

5.1 A uniform perturbation lemma

For $m, s, x, t \in [0, 1]$ define

$$f(m, s, x, t) = \frac{m - x - t}{D_x(m, s)D_t(m, s)}. \quad (5.1)$$

The elementary bound

$$\frac{3}{4} \leq D_u(m, s) \leq 2 \quad (0 \leq u, m, s \leq 1) \quad (5.2)$$

follows by minimizing $u^2 - mu$.

Lemma 5.1. *On the unit four-dimensional cube,*

$$|f_t| \leq \frac{304}{27} < 12, \quad |f_{tm}|, |f_{ts}| \leq \frac{4224}{81} < 64. \quad (5.3)$$

Consequently, if $\epsilon = 2^{-b}$ and

$$G_\infty(m, s) = \mathbb{E} \frac{m - X}{D_X(m, s)}, \quad (5.4)$$

then, uniformly for $m, s \in [0, 1]$,

$$|G_b - G_\infty| \leq 12\epsilon, \quad |(G_b)_m - (G_\infty)_m| \leq 64\epsilon, \quad |(G_b)_s - (G_\infty)_s| \leq 64\epsilon. \quad (5.5)$$

Proof. Put $Q = (D_x D_t)^{-1}$ and $N = m - x - t$. Then $|N| \leq 2$ and, using (5.2),

$$|Q| \leq \frac{16}{9}, \quad |Q_m|, |Q_s|, |Q_t| \leq \frac{128}{27}, \quad |Q_{tm}| \leq \frac{1728}{81}, \quad |Q_{ts}| \leq \frac{1920}{81}.$$

Here $|(D_t)_t| \leq 2$, $|(D_u)_m|, |(D_u)_s| \leq 1$, $|(D_t)_{tm}| \leq 1$, and $|(D_t)_{ts}| \leq 2$ give the bounds directly by the product rule. Thus

$$|f_t| \leq |Q| + 2|Q_t| \leq \frac{304}{27},$$

$$|f_{tm}| \leq |Q_m| + |Q_t| + 2|Q_{tm}| \leq \frac{4224}{81}, \quad |f_{ts}| \leq |Q_s| + 2|Q_{ts}| \leq \frac{4224}{81}.$$

The fundamental theorem of calculus in t compares $t = XV$ with $t = 0$. Since $0 \leq XV \leq V$ and $\mathbb{E}V = 2^{-b}$, expectation yields (5.5). The bounded derivatives also justify passing m, s derivatives through the expectation. $\square$

5.2 A variance gap that is uniform in radius

Lemma 5.2. *For $0 \leq m, s \leq 1$,*

$$(G_\infty)_m = \mathbb{E} D_X^{-2} \geq \frac{1}{4}, \quad (G_\infty)_s = \mathbb{E} \frac{X(X-m)^2}{D_X^2} \geq \frac{\Delta(a)}{4}. \quad (5.6)$$

Also

$$G_\infty(1, s) \geq \frac{1 - (2/3)^a}{2}, \quad 0 < \Delta(a) < 1 - (2/3)^a < 1. \quad (5.7)$$

Proof. Differentiation of (5.4) gives the two identities in (5.6). By $D_X \leq 2$ and completing the square in m ,

$$\begin{aligned} (G_\infty)_s &\geq \frac{1}{4} \mathbb{E}[X(X-m)^2] \\ &\geq \frac{1}{4} \left(\mathbb{E}X^3 - \frac{(\mathbb{E}X^2)^2}{\mathbb{E}X} \right) = \frac{1}{4} ((2/5)^a - (3/8)^a). \end{aligned}$$

The difference is strictly positive because $2/5 > 3/8$. At $m = 1$, $1 - X \geq 0$ and $D_X \leq 2$, proving the endpoint bound. Finally

$$\Delta(a) = (2/5)^a [1 - (15/16)^a] < 1 - (2/3)^a.$$

□

Proof of Theorem 1.2. Assume (1.8). Since $\Delta(a) < 1$, this implies $b > 8$ and hence $a + b > 1$, so the inherited uniqueness theorem (1.2) applies. Equations (5.5) and (5.6) give

$$(G_b)_m > \frac{1}{4} - \frac{\Delta(a)}{4} > 0, \quad (G_b)_s > \frac{\Delta(a)}{4} - \frac{\Delta(a)}{4} = 0.$$

Moreover $G_b(0, s) < 0$ directly from (2.8), whereas

$$G_b(1, s) > \frac{1 - (2/3)^a}{2} - \frac{3\Delta(a)}{64} > 0.$$

There is therefore a simple root in $0 < m < 1$, and it is the unique upper-semicircle root. Its derivative is $m'(s) = -(G_b)_s / (G_b)_m < 0$, which proves the asserted direction.

Under $\epsilon \leq \Delta(a)/512$, the same estimates give $(G_b)_s \geq \Delta(a)/8$ and

$$(G_b)_m \leq \frac{16}{9} + \frac{\Delta(a)}{8} < 2.$$

Thus $m'(s) \leq -\Delta(a)/16$. Equation (1.3) proves (1.9). □

Corollary 5.3 (A compact outer-order interval). *Fix $0 < \alpha \leq A < \infty$. If*

$$2^{-b} < \frac{1}{256} \left(\frac{2}{5} \right)^A \left[1 - \left(\frac{15}{16} \right)^\alpha \right],$$

then strict global decrease holds simultaneously for every $a \in [\alpha, A]$.

Proof. The displayed numerator is a lower bound for $\Delta(a) = (2/5)^a [1 - (15/16)^a]$ throughout that interval. □

The perturbation theorem supplies more than pointwise convergence as $b \rightarrow \infty$. It controls the zero equation and both of its relevant derivatives uniformly in radius, and gives a positive lower gap before the perturbation is made. Without these two properties, a strict sign in the limit need not persist on the entire interval.

6 An exact cubic-moment criterion for radial motion

The previous arguments prove two sign regimes. The following identity isolates the obstruction in the remaining cases. It is stated on $a, b > 0, a + b > 1$, where (1.2) supplies the required simple root and sign. The algebra also holds at any other simple root satisfying the same geometric hypotheses.

Write $\widehat{\eta}(s) = \eta_{a,b}(\sqrt{s}) = m(s)/2$, and, at its zero, put

$$A = 1 - s\widehat{\eta}^2 > 0, \quad D = 1 + s(u^2 - 2\widehat{\eta}u), \quad M_j = \int (u - \widehat{\eta})^j D^{-3} d\mu(u). \quad (6.1)$$

The inequality $A > 0$ follows from $0 < \widehat{\eta} < 1$ and $s \leq 1$.

Theorem 6.1 (Weighted cubic-moment velocity). *At the angular zero,*

$$AM_1 + sM_3 = 0 \quad (6.2)$$

and

$$\widehat{\eta}'(s) = -\frac{2M_3}{A^2M_0 - 3sAM_2 + 4s^2\widehat{\eta}M_3}. \quad (6.3)$$

The denominator is strictly positive. Consequently the sign of $\eta'_{a,b}(\rho)$ is the opposite of the sign of M_3 .

Proof. Set

$$\mathcal{G}(\eta, s) = \int (\eta - u)[1 + s(u^2 - 2\eta u)]^{-2} d\mu(u).$$

Equation (2.7) gives $P(2\eta, s) = 2\mathcal{G}(\eta, s)$. Since $D = A + s(u - \eta)^2$, the equation $\mathcal{G} = 0$ is exactly (6.2). Holding η fixed when differentiating in s ,

$$\mathcal{G}_s = 2(M_3 - \eta^2 M_1) = \frac{2M_3}{A}.$$

Similarly,

$$\mathcal{G}_\eta = AM_0 - 3sM_2 - 4s\eta M_1 = AM_0 - 3sM_2 + \frac{4s^2\eta M_3}{A}.$$

Implicit differentiation gives (6.3). At the inherited angular zero, the angular derivative of the imaginary part is negative, while $\partial_\theta m = -2 \sin \theta / \rho < 0$. Equation (2.7) therefore gives $P_m > 0$. As $\mathcal{G}_\eta = P_m(2\eta, s) > 0$, the denominator in (6.3) is $A\mathcal{G}_\eta > 0$. Finally $\eta'(\rho) = 2\rho\widehat{\eta}'(s)$, which does not change sign. $\square$

Corollary 6.2 (A precise reformulation of the remaining conjecture). *For integer $a \geq 2$, global normalized-radius decrease is equivalent to positivity of*

$$\int_0^1 \frac{(u - \eta_{a,b}(\rho))^3}{[1 + \rho^2(u^2 - 2\eta_{a,b}(\rho)u)]^3} d\mu_{a,b}(u)$$

at every $0 < \rho \leq 1$. The integral is strictly positive in both regimes of Theorems 1.1 and 1.2.

This moment is centered at the root parameter, not necessarily at the mean of the reweighted measure. Calling it a skewness criterion is useful shorthand, but it is not the usual normalized statistical skewness. Positivity of μ alone does not determine its sign.

7 Positive binomial identities and exact remainders

Let $g_{a,b} = \text{Im } F_{a,b}(i)$. The squared-resolvent representation makes a particularly simple family of alternating finite sums positive.

Theorem 7.1 (Positive Gaussian binomial transform). *For $a, b > 0$ and every integer $k \geq 0$, define*

$$\mathcal{B}_k(a, b) = \sum_{j=0}^k (-1)^j \binom{k}{j} \frac{H_{2j+2}^{(b)}}{(2j+2)(2j+3)^a}. \quad (7.1)$$

Then $\mathcal{B}_k(a, b) > 0$, and

$$-g_{a,b} = \sum_{k=0}^{\infty} \frac{k+1}{2^{k+1}} \mathcal{B}_k(a, b). \quad (7.2)$$

For the partial sum S_N over $0 \leq k < N$, the remainder is exactly

$$-g_{a,b} - S_N = 2^{-N} \int_0^1 \frac{u(1-u^2)^N [(N+2) + Nu^2]}{(1+u^2)^2} d\mu(u), \quad (7.3)$$

and hence, for every $N \geq 1$,

$$0 < -g_{a,b} - S_N < \frac{N+2}{2^{N+1}} \frac{H_2^{(b)}}{3^a}. \quad (7.4)$$

Proof. By (2.4),

$$\mathcal{B}_k(a, b) = \int u(1-u^2)^k d\mu(u) > 0, \quad -g_{a,b} = 2 \int \frac{u}{(1+u^2)^2} d\mu(u).$$

Set $q = (1-u^2)/2 \in [0, 1/2]$. The identity $(1-q)^{-2} = \sum_{k \geq 0} (k+1)q^k$ gives (7.2); all terms are nonnegative, so monotone convergence justifies integration. The finite algebraic remainder

$$\sum_{k=N}^{\infty} (k+1)q^k = q^N \frac{(N+1) - Nq}{(1-q)^2}$$

gives (7.3) after substitution. For $v \in [0, 1]$, $[(N+2) + Nv]/(1+v)^2 \leq N+2$. Since $\int u d\mu = c_3/2$, (7.4) follows, with strictness because μ has positive mass in the open interval. $\square$

This theorem concerns a different transformation from the manuscript's Euler transform of odd harmonic coefficients. The one-sided remainder and the positivity of each $\mathcal{B}_k$ are useful structural facts; no claim is made that the bound $O(N2^{-N})$ supersedes the sharper existing $O(2^{-N})$ evaluator [7].

7.1 Three explicit specializations

Corollary 7.2 (An identity for $\pi \log 2$).

$$\pi \log 2 = 4 \sum_{k=0}^{\infty} \frac{k+1}{2^k} \sum_{j=0}^k (-1)^j \binom{k}{j} \frac{H_{2j+2}}{(2j+2)(2j+3)}. \quad (7.5)$$

Every inner finite sum is positive, and the error after N outer terms is strictly between zero and $(N+2)2^{1-N}$.

Proof. The classical identity $F_{1,1}(z) = \frac{1}{2} \log^2(1-z)$ follows directly by differentiating its series and matching the value at zero. At $z = i$, $\log(1-i) = \frac{1}{2} \log 2 - i\pi/4$, whence $-g_{1,1} = \pi \log 2/8$. Apply [Theorem 7.1](#) with $c_3 = H_2/3 = 1/2$ and multiply the value and error bound by eight. $\square$

Define $\beta(t) = \text{Im Li}_t(i)$, the Dirichlet beta function with analytic continuation when necessary. Taking the endpoint limits in [Theorem 7.1](#) gives two further identities.

Corollary 7.3 (Beta differences). *For $a > 0$,*

$$\beta(a) - \beta(a-1) = \frac{1}{2} \sum_{k=0}^{\infty} \frac{k+1}{2^k} \sum_{j=0}^k (-1)^j \binom{k}{j} (2j+3)^{-a}, \quad (7.6)$$

$$1 - \beta(a) = \frac{1}{2} \sum_{k=0}^{\infty} \frac{k+1}{2^k} \sum_{j=0}^k (-1)^j \binom{k}{j} \frac{1}{(2j+2)(2j+3)^a}. \quad (7.7)$$

Every inner sum in these formulas is strictly positive. In particular $a = 2$ in (7.6) gives an explicit positive series for Catalan's constant minus $\pi/4$.

Proof. As $b \downarrow 0$, $V \rightarrow 1$ in probability in (2.3); as $b \rightarrow \infty$, $V \rightarrow 0$ in probability. All relevant kernels at $z = i$ are bounded and continuous. The limits are respectively $F_{a,0}(z) = \text{Li}_{a-1}(z) - \text{Li}_a(z)$ and $F_{a,\infty}(z) = \text{Li}_a(z) - z$. Their positive squared-resolvent measures are, respectively, c_2 times the law of X and c_2 times the mixture of the uniform laws on $[0, X]$. Both have full support. Their moments are the limits of (2.4), which give the two displayed inner sums. Apply the same proof as [Theorem 7.1](#) to these limiting measures. This avoids relying on an unjustified interchange of two infinite sums. $\square$

7.2 The generating function of the positive moments

For $|t| < 1$, another useful exact identity is

$$\sum_{k=0}^{\infty} \mathcal{B}_k(a, b) t^k = \int_0^1 \frac{u}{1-t+tu^2} d\mu(u). \quad (7.8)$$

The geometric expansion is uniformly convergent on compact subdisks. For $0 < t < 1$ the right-hand side is positive. More precisely, $\frac{1}{2} \frac{d}{dt} (t \sum_k \mathcal{B}_k t^k) \big|_{t=1/2} = -g_{a,b}$, which recovers (7.2). More generally, all derivatives in $t \in [0, 1)$ are positive, providing a generating-function form of the finite-sum positivity.

8 Reproducible exact computation

All acceptance decisions in the supplied verifiers use Python's standard-library `Fraction`. The analytic arguments, not finite sampling, transfer those decisions to statements about the infinite functions and continuous parameter ranges.

8.1 Global certificate

The verifier code `verify_global.py` checks the ten coefficients in [table 1](#), the root-strip and radial-tail inequalities, the rational comparisons extending the core to real $a \geq 8$, and the eight integer thresholds for [Theorem 1.2](#). The finite sums use cutoff 100. Representative exact outward bounds printed by the verifier are

Certified quantity	Outward decimal information
Upper bound for B_8	< 0.302408858414858
Localization margin	> 0.022510694406543
Upper bound for R_8	< 0.388094585988858
Low-mode contribution divided by c_5	$> 2/5$
Full P_s/c_5 at the root	$> 1/100$

These decimals are not asserted to be optimal constants.

8.2 Independent angular-root certificates

There is an exponentially convergent rational evaluator for [\(2.7\)](#). On $0 \leq m \leq 1/3, 0 \leq s \leq 1$, define

$$E(u) = 1 - \frac{2}{3}D_u = \frac{1}{3} + \frac{2sm}{3}u - \frac{2s}{3}u^2. \quad (8.1)$$

Since $35/36 \leq D_u \leq 2$, $|E(u)| \leq 19/54 < 3/8$. Thus

$$P_N(m, s) = \frac{4}{9} \sum_{k=0}^{N-1} (k+1) \int (m-2u)E(u)^k d\mu(u) \quad (8.2)$$

is computable from finitely many rational moments when a, b, m, s are rational with integer a, b . Its error is bounded by

$$|P - P_N| \leq \frac{8}{9}c_2 \left(\frac{3}{8}\right)^N \frac{(N+1) - 3N/8}{(1-3/8)^2}. \quad (8.3)$$

Indeed $|m-2u| \leq 2$, and the differentiated geometric tail used in [Theorem 7.1](#) bounds the absolute series. This estimate is valid even though E can be negative.

The file `data/root_brackets.json` records candidate rational endpoints. Their method of discovery has no evidentiary role: the verifier accepts a bracket only when the upper error bound for P at the lower endpoint is negative and the lower error bound at the upper endpoint is positive. Monotonicity in m then certifies the unique root. The replay uses $N = 36$ and certifies the following enclosures for $\eta = m/2$; the displayed endpoints are rounded outward.

a	b	ρ	Lower bound for η	Upper bound for η
8	1	1/2	0.029174544600	0.029174544751
8	1	1	0.028924151300	0.028924151451
8	2	1/2	0.024321334850	0.024321335001
8	2	1	0.024137504900	0.024137505051
9	1	1/2	0.019469017900	0.019469018051
9	1	1	0.019354467350	0.019354467501
12	1	1/2	0.005777259000	0.005777259151
12	1	1	0.005767707400	0.005767707551

These checks illustrate and independently test the normalization and root evaluator. They are not used as a substitute for the uniform proof.

8.3 Gaussian enclosures and an elementary control

The script code/verify_identities.py computes 80 terms of (7.2) for nine integer parameter pairs. It verifies 720 strictly positive rational values of $\mathcal{B}_k$, and separately checks 100 polynomial geometric-remainder identities. The following are exact outward enclosures for $-g_{a,b}$:

a	b	Lower bound	Upper bound
1	1	0.272198261287950266	0.272198261287950267
1	2	0.251642673723681485	0.251642673723681486
2	1	0.113648917599903643	0.113648917599903644
2	2	0.101022920883400012	0.101022920883400013
3	4	0.032901765615888080	0.032901765615888081
8	1	0.000223662993166255	0.000223662993166256
8	2	0.000187104280701129	0.000187104280701130
9	4	0.000053452648646007	0.000053452648646008
12	1	0.000002814149505465	0.000002814149505466

For $(a, b) = (1, 1)$ an independent interval for $\pi \log 2/8$, obtained from rational arctangent and logarithm series, lies strictly inside the unrounded binomial enclosure. This checks the scale and sign against an elementary special value without relying on a floating-point polylogarithm implementation. Details are in [section B](#).

9 Source audit, corrections, and integration boundaries

9.1 What should change in the manuscript

The final remark of the local-radius section should no longer leave *all* integer outer orders in the global conjectural column. It can say that full-radius decrease is proved for every real $a \geq 8$, and for every $a > 0$ under (1.8), while preserving the remaining cases as open. The left-endpoint limit theorem in the Cartesian-motion section should be followed by the quantitative finite- b transfer rather than being replaced or presented as new.

The new binomial identities can be added alongside positive-kernel identities. Their computational status should remain distinct from the sharp Euler theorem. The supplied integration/

files use namespaced labels and provide an additive insertion, not an automatic rewrite of the canonical manuscript.

9.2 A small existing notation correction

The proof in `05-real-positive-kernel.tex` of the Hurwitz–polylogarithm transport theorem contains the phrases “with $q = n$ ” and “boundedness of h ”. These are inconsistent with the remainder kernel notation $q(t)$ used in the same formulas. The intended substitution is the positive scalar argument of the Hurwitz formula equal to n ; the bounded kernel in the displayed integral is $q(t)$. A safe wording is: “Apply the Hurwitz difference formula at the positive integer argument n ; use boundedness of $q(t)$ near zero and its bounded large- t behavior.” This is a notation repair, not a refutation of the transport identity. It should be checked against the surrounding notation at integration time. These editorial repairs should be deduplicated against incoming audit notes during integration; no claim of first discovery is made.

9.3 Non-errors that must not become errors during integration

The inspected local theorem is correctly restricted to a neighborhood of the origin. We found no new fatal flaw in that proof. A local coefficient sign must not be cited as a global radial sign. Likewise, the Cartesian quantity $\rho \cos \theta$ and the normalized quantity $\cos \theta / \rho$ have different derivatives: the manuscript’s outward-abscissa theorem does not settle the normalized conjecture.

The incoming fractional bifurcation article has a global statement in the inner-order parameter *along a quadratic threshold*, but its turning-point geometry is local in radius. Those quantifiers should not be combined with the all-radius quantifiers of the present paper. The new large- a theorem does not erase the fractional counterexamples.

Finally, the positive squared-resolvent measure is not the manuscript’s signed first-resolvent measure, and analytic positivity is not an arithmetic depth-reduction certificate. The S_6 and surviving S_8 candidates keep their existing status. We do not present earlier all-depth oscillation, the sharp Euler constant, or the local quartic transition as newly solved here.

10 Further research questions and proposed directions

10.1 The six remaining integer strips

The most concrete next target is

Conjecture 10.1 (Remaining integer normalized-radius problem). For $a \in \{2, 3, 4, 5, 6, 7\}$ and every $b > 0$, $\eta'_{a,b}(\rho) < 0$ on $0 < \rho \leq 1$.

By [Theorem 1.2](#), only $0 < b < B_a$ from the threshold table remains. The squared-resolvent evaluator is uniform in radius and avoids slowly converging unit-circle derivatives. This suggests a genuine computer-assisted proof strategy: cover each bounded (b, s) region, isolate the root in m , and certify positivity of P_s there using interval moments and analytic tails. Such a proof would require outward bounds for the noninteger powers in $H_n^{(b)}$, and a treatment of the edges $b \downarrow 0$ and $s \downarrow 0$. No such complete parameter-covering certificate is included here.

10.2 The first outer order

The manuscript predicts decrease for $a = 1, b > 1$, increase for $a = 1, 0 < b < 1$, and the constant $\eta = 1/2$ at $(1, 1)$. Our transfer theorem proves the decreasing claim only for $b \geq 14$. The range $1 < b < 14$ remains a focused target. An integral comparison against the exact logarithmic solution may be more efficient there than a mode-by-mode estimate.

10.3 How small can the uniform outer threshold be?

The number eight is an explicit sufficient threshold of this proof, not a demonstrated phase boundary. Retaining favorable signs from more even and odd modes, or certifying the core in a smaller actual parameter region $y = x^{\log_2 3}$ rather than $x^2 \leq y \leq x$, may lower it. A substantially different question asks for the least real A such that global decrease holds for all $a \geq A, b > 0$. Fractional turning-point phenomena make it inappropriate to infer the answer from the local $a \geq 2$ result alone.

10.4 Sharp stability in the inner order

The constants 12, 64, and 256 arise from deliberately uniform kernel bounds. Determine the least sufficient $B(a)$, or an asymptotically sharp $B(a)$ as $a \downarrow 0$ and $a \rightarrow \infty$. The proof already isolates the useful variance gap $\Delta(a) = \mathbb{E}X^3 - (\mathbb{E}X^2)^2/\mathbb{E}X$; retaining $\mathbb{E}(XV)$ rather than bounding it by $\mathbb{E}V$, and exploiting the root equation, should improve the perturbation threshold. These observations are proposals, not claims of a sharp asymptotic formula.

10.5 A structural cubic-moment theorem

The criterion in [Theorem 6.1](#) reduces radial motion to a sign of a reweighted third moment. Characterize positive measures μ for which that sign is positive at every root. A useful theorem should state additional shape or transform hypotheses, because bare positivity cannot suffice. The harmonic moment family offers an explicit test class with known fractional changes of sign.

10.6 Further identities and arithmetic certificates

Equation (7.8) suggests specializations at rational t , derivatives of arbitrary order, and cyclotomic analogues of the positive Gaussian transform. Their analytic convergence and remainder bounds are accessible from the same measure. The separate question is whether any such identity yields a finite, meaningful reduction in the manuscript's arithmetic period basis. The S_6, S_8 targets require that second step; an accelerated positive infinite series is not itself the missing finite arithmetic certificate.

10.7 Deeper strict polylogarithms

The incoming all-depth report controls several angular roots for integer leading indices. It is natural to ask which normalized Cartesian displacements of those roots have uniform radial signs, and whether positive higher-resolvent measures lead to analogous central-moment identities. Multiple-star sums and strict sums must remain separate, as must the number of angular roots and the monotonicity of an individual normalized branch.

A Finite certificates and analytic tail bookkeeping

Let $N = 100$ and $M = M_8$. The following are rational upper bounds for the sums in (3.7):

$$\begin{aligned}\bar{B} &= \sum_{n=3}^N n(n-1)(2/n)^8 + \frac{2^8}{5N^5}, \\ \bar{O} &= \sum_{\substack{3 \leq n \leq N \\ n \text{ odd}}} \frac{n-1}{2}(3/n)^8 + \frac{3^8}{12N^6}, \\ \bar{E} &= \sum_{\substack{4 \leq n \leq N \\ n \text{ even}}} n(n-1)(2/n)^8 + \frac{2^8}{5N^5}.\end{aligned}$$

For example, $n(n-1)n^{-8} \leq n^{-6}$ and $\sum_{n>N} n^{-6} \leq \int_N^\infty t^{-6} dt = 1/(5N^5)$. For the odd sum, use $(n-1)n^{-8}/2 \leq n^{-7}/2$ and bound the remaining odd indices by *all* indices; this is deliberately conservative. The same method bounds the radial sum by

$$\begin{aligned}\bar{R} &= \sum_{n=7}^N \frac{n-1}{4}(5/n)^8 B_n \\ &\quad + \frac{5^8}{4} \left(\frac{101}{200} + \frac{11M}{40} \right) \frac{1}{5N^5} + \frac{5^8 101M}{1600} \frac{1}{4N^4}.\end{aligned}\tag{A.1}$$

Indeed the odd summand is bounded by $5^8(101/200 + 11M/40)n^{-6}/4$, and the even summand by $5^8 101Mn^{-5}/1600$.

The verifier performs the exact rational comparisons

$$\bar{B} < \frac{1}{3}, \quad C \left(1 - \frac{101}{200} \bar{E} \right) - \frac{101}{100} \bar{O} > \frac{1}{50}, \quad \bar{R} < \frac{39}{100}.$$

For a polynomial $p(x) = \sum_{k=0}^d p_k x^k$, the degree- d Bernstein coefficients used in table 1 are

$$b_j = \sum_{k=0}^j p_k \frac{\binom{j}{k}}{\binom{d}{k}}.$$

The code computes these from (4.3) over $\mathbb{Q}$, compares them with the displayed integers divided by the common denominator, and checks that every coefficient is positive. The Bernstein basis functions are nonnegative and sum to one on $[0, 1]$, so these finite signs imply positivity on the full interval.

There is no sampling in a or b in this proof: the exponential comparison proves the extension from $a = 8$ to all real $a \geq 8$, and the affine/Bernstein argument covers every $b > 0$ through $x^2 \leq y \leq x$. Only the elementary infinite sums are truncated, with the rigorous integral tails displayed above.

B Independent control of the elementary special value

For rational $0 < x < 1$, consecutive partial sums of

$$\arctan x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{2k+1}$$

bracket the exact value by the alternating-series theorem. The verifier uses 100 terms at $x = 1/5$ and 35 at $x = 1/239$, together with Machin's identity

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239).$$

For a direct check of the identity, the tangent addition formula gives $\tan(4 \arctan(1/5)) = 120/119$ and $\tan(4 \arctan(1/5) - \arctan(1/239)) = 1$; the angle is in $(0, \pi/2)$.

For $\log 2$, use

$$\log 2 = 2 \sum_{k=0}^{\infty} \frac{(1/3)^{2k+1}}{2k+1}.$$

After L terms the positive remainder is at most

$$\frac{2(1/3)^{2L+1}}{(2L+1)(1-1/9)}.$$

The verifier takes $L = 110$, multiplies the positive intervals for π and $\log 2$, divides by eight, and verifies strict containment inside the rational $(1, 1)$ Gaussian enclosure. Every endpoint in this independent calculation is rational.

C Reproduction, dependency map, and integration

From the package root, with Python 3.10 or newer and without optimized mode, run

```
python code/run_all.py
```

The individual scripts are

```
python code/verify_global.py
python code/verify_roots.py
python code/verify_identities.py
```

They regenerate their receipts under `data/`. Run them on a copy when preservation of delivered evidence bytes matters. No third-party Python dependency is needed. To rebuild the PDF with a standard TeX installation, run `build.sh`, or run `pdflatexarticle.tex` twice. The bibliography is included in the source, so no bibliography processor or network access is required.

The proof dependencies are deliberately limited. The $a \geq 8$ theorem uses the inherited positive kernel but proves its own root localization and uniqueness. The finite- b transfer uses the inherited uniqueness result only to identify the constructed $0 < m < 1$ root as the sole upper angular zero. The cubic criterion uses the inherited angular-derivative sign. The binomial identities depend only on the positive squared resolvent, not on the radial theorem. The exact root and Gaussian brackets test concrete values and normalizations independently of the global parameter argument.

The package is additive and makes no remote repository changes. Suggested placement is a new report directory under `Analysis/Polylogarithms/docs/reports/`, with the namespaced excerpt from `integration/` inserted in the real-order chapter after the local-radius material. The status ledger should record the two proved global regimes and preserve the unresolved strips, the first-order cases not covered here, and the arithmetic conjectures separately.

References

- [1] ProveIt Contributors, *A proved local part of the normalized-radius conjecture*, Analysis/Polylogarithms/docs/manuscript/chapters/05-real-local-radius.tex, Git blob 30e3e31944be5bc0d03a8485386dd1608c075507. Repository <https://github.com/VladimirReshetnikov/ProveIt>, accessed 10 October 2026.
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